Get Started

What is 

R is a programming language strongly oriented towards statistics, data management and visualization.

It was created in 1993 by **Ross Ihaka** and **Robert Gentleman**.

It is completely **open-source** and **free** software, constantly evolving and changing.

Why 

A word on open-source

A piece of software is called open-source when its **source code is available** to everyone to be **modified**, **updated** and **inspected**.

R is **both open-source and free**, and it has an extremely active community, as often happens with open-source projects and programming languages in general.

Who are the competitors?

R's main "competitor" is certainly **Python**, which offers an equally powerful, well-developed and active environment.

It is not easy (and maybe not even possible) to say which one is better.

In any case, once you have learned R, learning Python will be very easy.

Who are NOT the competitors?

In the field of statistics there are several non-open-source, paid software packages such as:

- Statistica
- SPSS
- STATA
- SAS

Who are NOT the competitors?

They are excellent pieces of software, but:

- They do not provide transferable skills
- You are tied to a specific environment
- Licenses can be very expensive
- The community is not as active (not open-source)

What are the alternatives?

There are some excellent open-source programs based on R, such as:

- Jamovi
- Jasp: many models, including advanced ones!

Learning a language as an investment

Learning a language like R lets you master a very powerful tool, but also learn to:

- Reason and solve problems with code
- Transfer what you have learned to other languages
- Always be autonomous and not tied to a specific environment
- Have a truly valuable skill

Learning R is hard…BUT it’s worth it!

Installing R and R-Studio

R and R-studio are two different pieces of software.




```
R version 4.2.3 (2023-03-15) -- "Shortstop Beagle"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

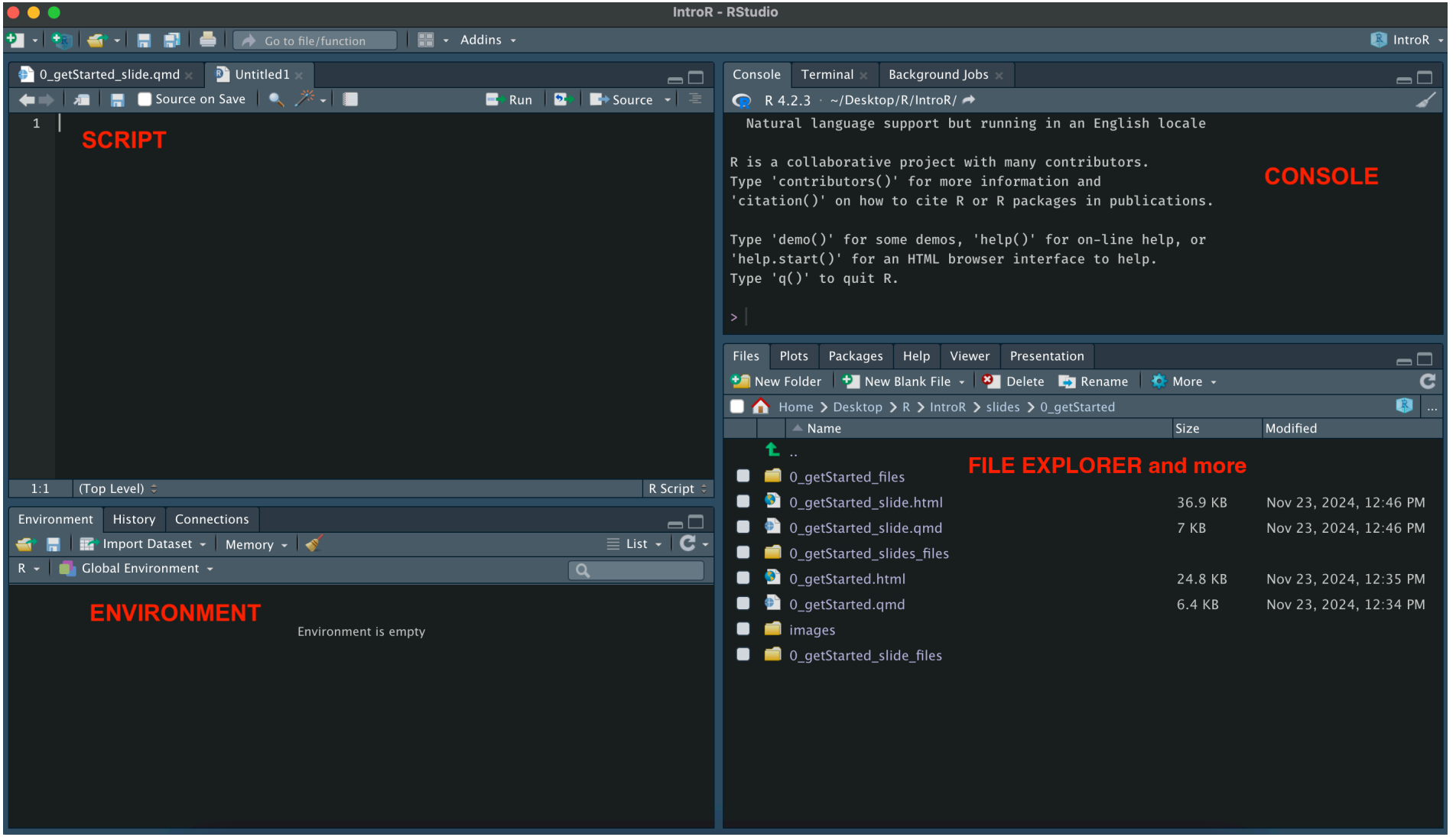
During startup - Warning messages:
1: Setting LC_CTYPE failed, using "C"
2: Setting LC_COLLATE failed, using "C"
3: Setting LC_TIME failed, using "C"
4: Setting LC_MESSAGES failed, using "C"
5: Setting LC_MONETARY failed, using "C"
[R.app GUI 1.79 (8198) x86_64-apple-darwin17.0]

WARNING: You're using a non-UTF8 locale, therefore only ASCII characters will work.
Please read R for Mac OS X FAQ (see Help) section 9 and adjust your system preferences accordingly.
[Workspace restored from /Users/tita/.RData]
[History restored from /Users/tita/.Rapp.history]

>
```

R-Studio

R-studio is a user-friendly interface that lets you use R.



Installing R and R-Studio

They must be installed separately, and the procedure depends on your operating system.

If you have not installed them yet, follow the procedure explained at this link: https://posit.co/download/rstudio-desktop/

Important things to know

1. Leave R and RStudio where they are: your computer installs them in the right place automatically
2. Create a folder named `introR`, on your Desktop or wherever you prefer: it will be our **working directory** for the entire course